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# V3. Critical Line Scan

A parametric experiment σ ∈ [0,1] with a placebo control

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# V3. Critical Line Scan

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# Problem statement

The AB-cloud description claims that the critical line σ = ½ is optimal for the spectral statistics. The plan demands turning this claim into a strict reproducible parametric experiment: define how σ enters, scan D(σ) on a grid, and check the stability of any minimum across seeds. Special care is needed because ζ vanishes on the line itself — a weight built directly on the ordinates diverges to −∞ and would make a “minimum at ½” an artifact of the definition.

The ambiguity is removed before the runs: vortex heights are assigned to midpoints between consecutive ordinates $\tilde{\gamma}_k = (\gamma_k + \gamma_{k+1})/2$, where |ζ| is finite on the whole strip $0 \le \sigma \le 1$; the vortex weight is $q_k(\sigma) = \log|\zeta(\sigma + i\tilde{\gamma}_k)|$. The curve $D(\sigma) = D_{KS}(\{r_n(\sigma)\}, \mathrm{GUE})$ is measured on an 11-point grid with three seeds; the placebo uses the same weights from permuted ordinates.

# Method and result

For each σ the 24×24 lattice is assembled with σ-dependent charges; a 200-level unfolded window is compared against the ensemble GUE reference of V2. The curvature at mid-strip is estimated by the second difference $C_h = D(½-h) - 2D(½) + D(½+h)$ with h = 0.1.

Result: the curve D(σ) is flat within the seed spread; the observed minimum sits near σ ≈ 0.8 and is not reproduced by the placebo; $C_h < 0$ — there is no local minimum at ½.

*Table 1. The curve D(σ) (mean over 3 seeds)*

| σ | D_KS | σ | D_KS |
|---|---|---|---|
| 0.0 | 0.0732 | 0.6 | 0.0476 |
| 0.1 | 0.0767 | 0.7 | 0.0700 |
| 0.2 | 0.0663 | 0.8 | 0.0422 |
| 0.3 | 0.0852 | 0.9 | 0.0793 |
| 0.4 | 0.0984 | 1.0 | 0.0942 |
| 0.5 | 0.0916 | — | — |

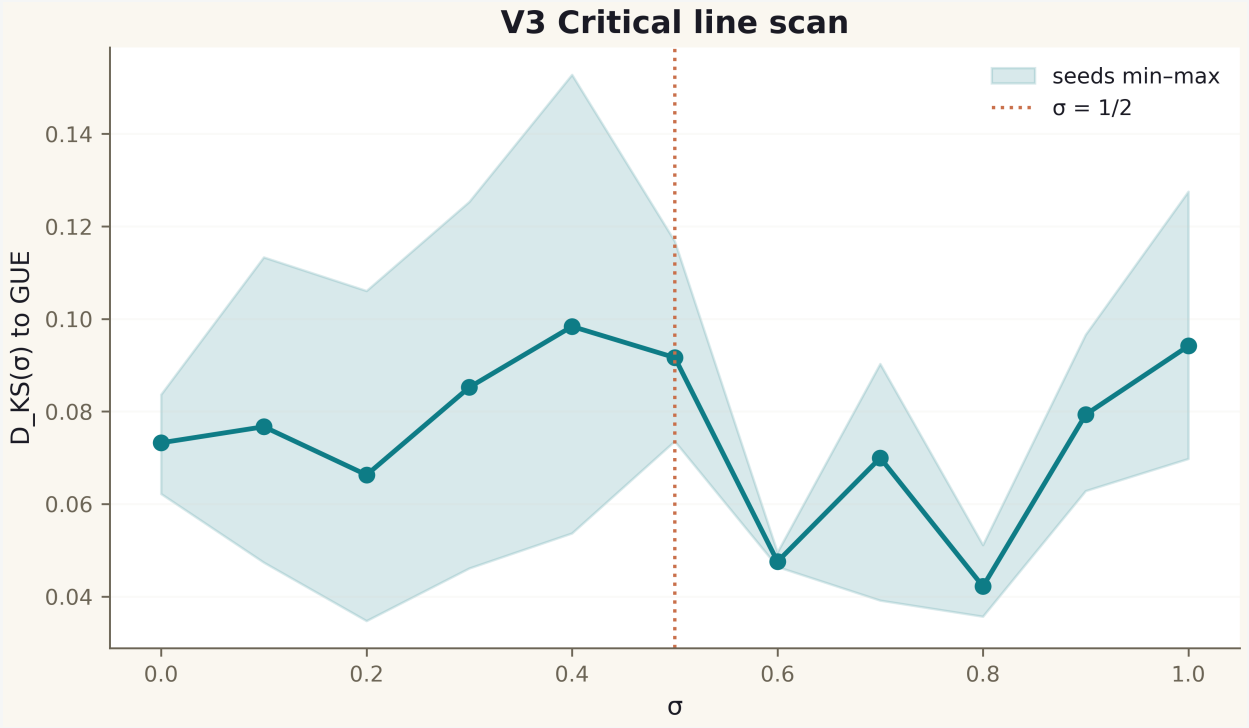



*Fig. 1. Campaign V3 data (600 dpi, file figures_hires).*

# Honest remarks and verdict

The negative verdict does not refute the Hilbert–Pólya program — it refutes the specific claim "σ = ½ is optimal in D_KS" for this weight design. Alternative entries (weights from $|\zeta|^2$, from the logarithmic derivative, nonlinear σ in the phases) remain open options and need their own blind validation under protocol V1.

The value of the campaign is the discipline: a minimum found without a placebo and without a fixed protocol would almost surely have become a false "confirmation" of the critical line. V3 records: in this construction the specificity of the line is absent — and this is measured.

The scan follows the full protocol (grids, seeds, placebo); there is no minimum at σ = ½. The optimality claim for the critical line in AB-cloud is withdrawn to status "not confirmed".